Student Activity

10 minutes

Task

Using the interactive pixel grid (HTML file), choose **two pixels** that are moderately far apart, then find:

1. The shortest path between them using each adjacency type: 4, 8, and m
2. All three distances: D_e, D_4, D_8 (origin: row=0, column=0)

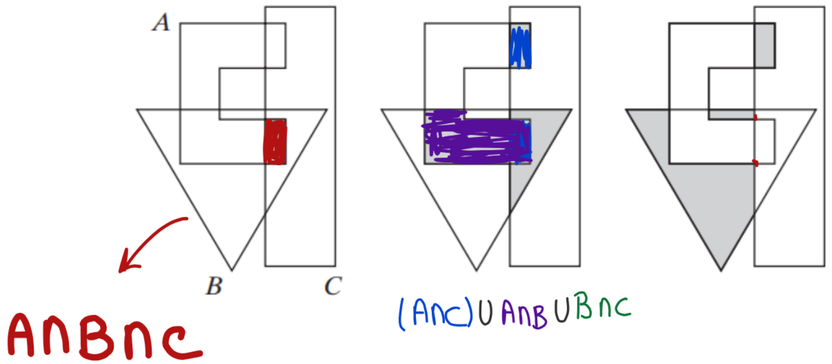
Submission

Draw the path on the grid (use Paint or equivalent).
Mark the **start pixel in blue** and the **end pixel in orange**.
Submit as a document.
Compare with the application.

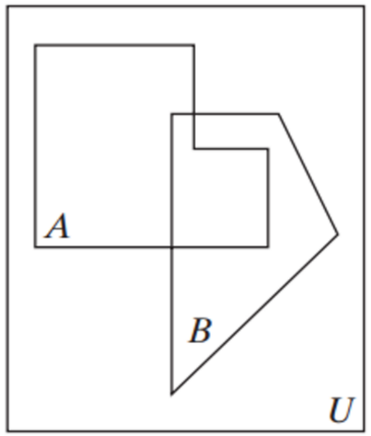
Textbook Problem 2.23 (3rd ed.) — Section Task

2.23 ★**(a)** With reference to Fig. 2.31, sketch the set $(A \cap B) \cup (A \cup B)^c$.

(b) Give expressions for the sets shown shaded in the following figure in terms of sets A, B, and C. The shaded areas in each figure constitute one set, so give one expression for each of the three figures.

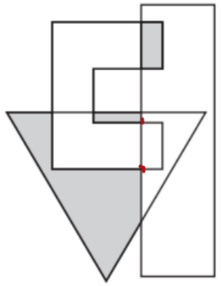


Problem 2.23 — set A



Task

Solve using Paint or any image editor. Draw the result directly on the image.



Problem 2.23 — solution/result image